# 10 Boolean logic

**1** State the logic gate each truth table represents.

**a**

| A | B | X |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

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**b**

| A | B | X |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

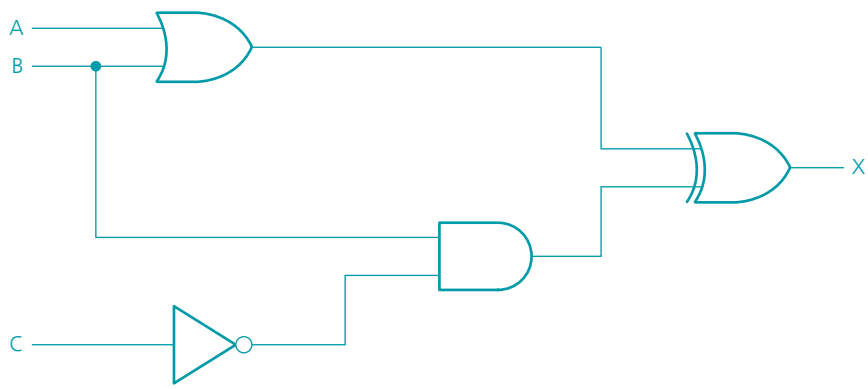
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**c**

| A | B | X |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

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**2** Complete the truth table for this logic circuit.




| A | B | C | Working area | X |
|---|---|---|---|---|
| 0 | 0 | 0 | | |
| 0 | 0 | 1 | | |
| 0 | 1 | 0 | | |
| 0 | 1 | 1 | | |
| 1 | 0 | 0 | | |
| 1 | 0 | 1 | | |
| 1 | 1 | 0 | | |
| 1 | 1 | 1 | | |

**3 a** Write the following statement as a logic expression:

X is 1 if A and B are on or if B is off and C is on

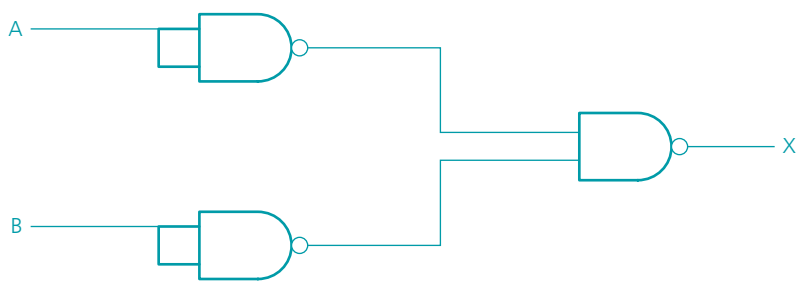
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**b** Draw the logic circuit to represent the statement in part **a**.




**4 a** Complete the truth table for the following logic circuit:

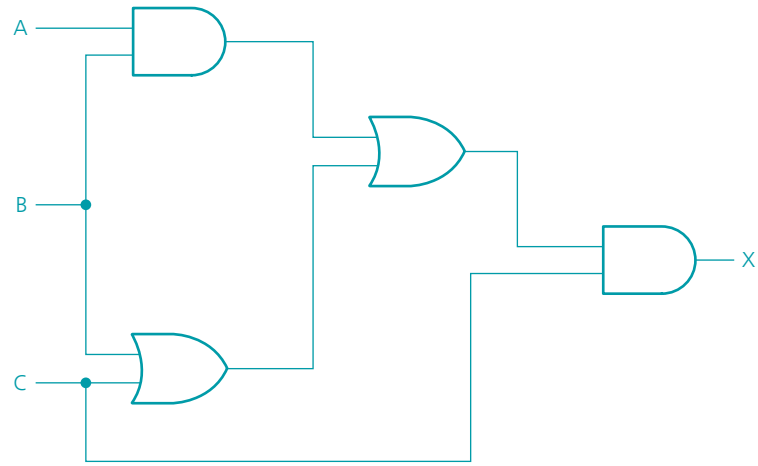



| A | B | Working area | X |
|---|---|---|---|
| 0 | 0 | | |
| 0 | 1 | | |
| 1 | 0 | | |
| 1 | 1 | | |

**b** Which single logic gate is represented by the truth table in part **a**?

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**5 a** Complete the truth table for the following logic circuit:



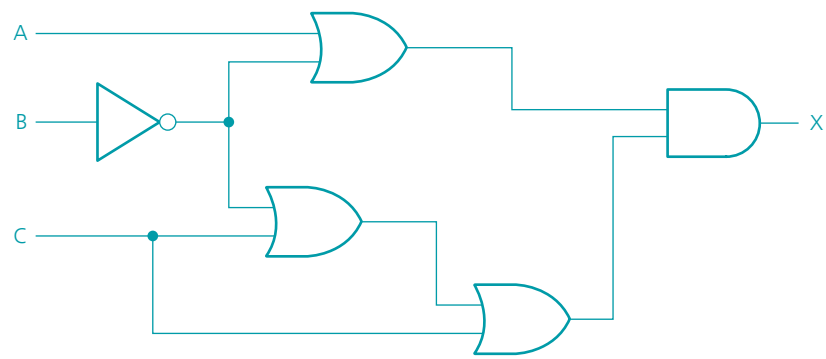

| A | B | C | Working area | X |
|---|---|---|---|---|
| 0 | 0 | 0 | | |
| 0 | 0 | 1 | | |
| 0 | 1 | 0 | | |
| 0 | 1 | 1 | | |
| 1 | 0 | 0 | | |
| 1 | 0 | 1 | | |
| 1 | 1 | 0 | | |
| 1 | 1 | 1 | | |

**b** How might the logic circuit in part **a** be simplified?

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**6 a** Complete the truth table for this logic circuit:



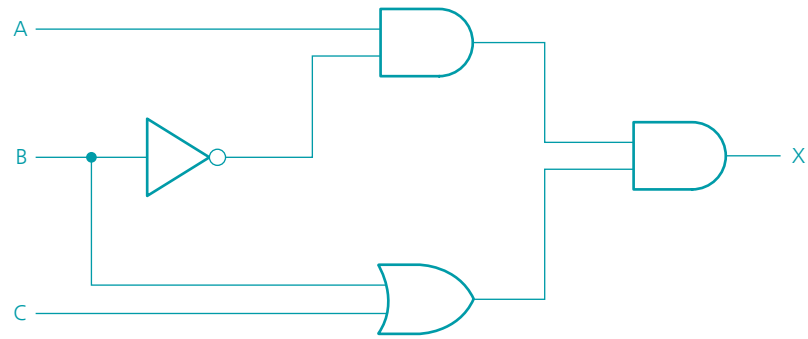

| A | B | C | Working area | X |
|---|---|---|---|---|
| 0 | 0 | 0 | | |
| 0 | 0 | 1 | | |
| 0 | 1 | 0 | | |
| 0 | 1 | 1 | | |
| 1 | 0 | 0 | | |
| 1 | 0 | 1 | | |
| 1 | 1 | 0 | | |
| 1 | 1 | 1 | | |

**b** What single logic gate could replace this logic circuit?




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**7 a** Write down the logic expression to represent the following logic circuit:




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**b** Draw the logic circuit which corresponds to the following truth table:

| A | B | C | X |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 |





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**8** A chemical process is monitored using a logic circuit. There are three inputs to the logic circuit representing the parameters being monitored in the chemical process. An alarm, X, will give an output value of 1 depending on certain conditions.

The following table describes the process conditions being monitored.

| Parameter description | Parameter | Binary value | Description of condition |
|---|---|---|---|
| Reaction temperature | T | 0 | Temperature > 120 °C |
| | | 1 | Temperature <= 120 °C |
| Pressure of CO gas evolved | P | 0 | Pressure > 2 bars |
| | | 1 | Pressure <= 2 bars |
| Acid concentration | A | 0 | Acid concentration > 4 moles |
| | | 1 | Acid concentration <= 4 moles |

An alarm, X, will generate the value 1 if:

either temperature > 120 °C and acid concentration <= 4 moles

or temperature <= 120 °C and gas pressure <= 2 bars

or acid concentration > 4 moles and gas pressure <= 2 bars

**a** Write the logic expression to represent the above system.

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**b** Draw the logic circuit to represent the above system.



**c** Complete the truth table to represent the above system.

| T | A | P | Working area | X |
|---|---|---|---|---|
| 0 | 0 | 0 | | |
| 0 | 0 | 1 | | |
| 0 | 1 | 0 | | |
| 0 | 1 | 1 | | |
| 1 | 0 | 0 | | |
| 1 | 0 | 1 | | |
| 1 | 1 | 0 | | |
| 1 | 1 | 1 | | |

**9** A safety system uses the input from three sensors A, B and C. The binary values from these sensors form the input to a logic circuit.

The output, X, from the logic circuit is 1 if:

either A is 1 and B is 1

or A is 0 and C is 1

or B is 0 and C is 1




**a** Draw the logic circuit to represent the above system.

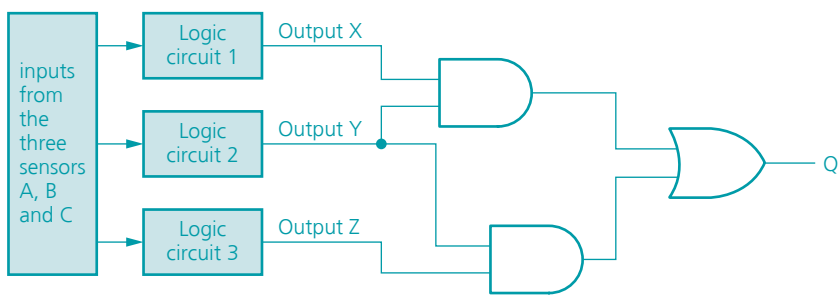


**b** Complete the truth table for the above system.

| A | B | C | Working area | X |
|---|---|---|---|---|
| 0 | 0 | 0 | | |
| 0 | 0 | 1 | | |
| 0 | 1 | 0 | | |
| 0 | 1 | 1 | | |
| 1 | 0 | 0 | | |
| 1 | 0 | 1 | | |
| 1 | 1 | 0 | | |
| 1 | 1 | 1 | | |

**c** For safety reasons, the sensors feed into three different logic circuits (numbered 1, 2 and 3) which produce the outputs X, Y and Z. These three outputs then form the inputs to another logic circuit which has the output Q.

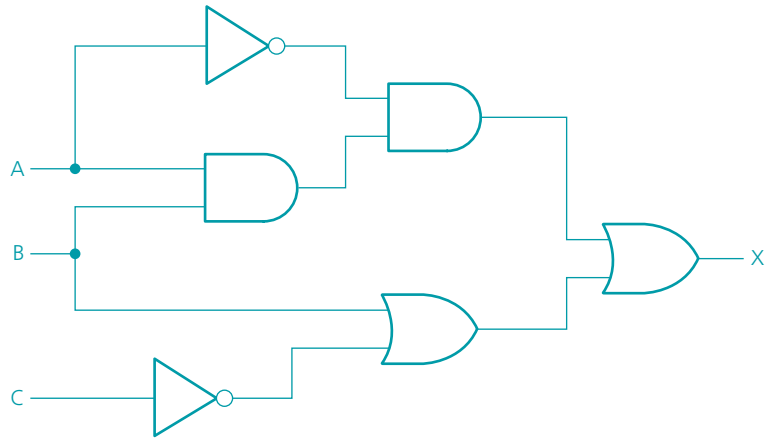
The three logic circuits are connected to this extra logic circuit:




Complete the truth table for this system.

| X | Y | Z | Working area | Q |
|---|---|---|---|---|
| 0 | 0 | 0 | | |
| 0 | 0 | 1 | | |
| 0 | 1 | 0 | | |
| 0 | 1 | 1 | | |
| 1 | 0 | 0 | | |
| 1 | 0 | 1 | | |
| 1 | 1 | 0 | | |
| 1 | 1 | 1 | | |

**10 a** Write a logic expression for the following logic circuit:




**b** Complete the truth table for the logic circuit.

| A | B | C | Working area | X |
|---|---|---|---|---|
| 0 | 0 | 0 | | |
| 0 | 0 | 1 | | |
| 0 | 1 | 0 | | |
| 0 | 1 | 1 | | |
| 1 | 0 | 0 | | |
| 1 | 0 | 1 | | |
| 1 | 1 | 0 | | |
| 1 | 1 | 1 | | |

**11 a** Write the logic expression for this truth table.

| A | B | X |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

**b** Draw the logic circuit which is represented by the truth table.

**12 a** Draw a logic circuit for this logic expression. Each logic gate must have a maximum of **two** inputs and the expression must not be simplified:

X = ((A NAND B) XOR C) OR (A NOR NOT C)




**b** Complete the truth table for the logic expression:

| A | B | C | Working area | X |
|---|---|---|---|---|
| 0 | 0 | 0 | | |
| 0 | 0 | 1 | | |
| 0 | 1 | 0 | | |
| 0 | 1 | 1 | | |
| 1 | 0 | 0 | | |
| 1 | 0 | 1 | | |
| 1 | 1 | 0 | | |
| 1 | 1 | 1 | | |